



Optimal Methodology Settings for Developing Revenue Prediction Models

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Abstract

Despite significant technological advancements, many business users lack the knowledge and expertise to effectively utilize business intelligence tools and machine learning techniques. This study aims to identify the optimal settings for developing predictive models tailored to SME business users. It focuses on determining the most robust machine learning model for revenue prediction and proposes deploying the model through an interactive financial dashboard, providing enhanced monitoring and decision-making capabilities. The Cross-Industry Standard Process for Data Mining (CRISP-DM) framework was enhanced by refining data understanding, preparation, model development and deployment using tools such as Power BI, JMP, and Python. For model development, machine learning algorithms such as XGBoost, random forest regression, ANN, and two hybrid models were tested. Additionally, an effective financial dashboard monitoring system was developed which allows stakeholders to monitor the company's revenue, costs, and financial health in a user-friendly manner to support informed decision-making. The findings show that by utilizing the enhanced CRISP-DM framework, optimal settings for data understanding, preparation, and model development were obtained. The “K-prototype + RFR” hybrid model exhibited superior performance with R^2 at 94% in revenue prediction, showing resilience to overfitting and effectively handling data noise. The developed financial dashboard monitoring system proved to be an intuitive and powerful tool that can assist stakeholders in efficiently tracking revenue and financial health. This study provides valuable insights into the application of advanced machine learning techniques for financial prediction and demonstrates the practical benefits of integrating predictive model results with user-friendly dashboard systems.

Keywords Predictive modeling · Revenue prediction · Enhanced CRISP-DM · Hybrid models · Financial monitoring

1 Introduction

With technological advancements and digital transformation, businesses possess massive amounts of data continuously generated throughout the business process. When utilizing the data correctly, the company can make informed and evidence-based decisions [5, 9]. One area in which the company can benefit from utilizing such data is financial planning. While Mitra et al. [19] argue that managers and decision-makers typically rely on their perception, intuition, and experience to make sales predictions, they highlight that this approach often leads to inconsistency in achieving accurate predictions. It is crucial for a business to draw accurate sales forecasts, as it assists business development and improves business strategies [31, 36].

An effective way of creating sales and revenue forecast models with big data is by utilizing machine learning techniques [32], as cited in [19]. Machine learning supports drawing data-driven decisions by producing predictions (Romero, Ventura, & discovery, 2020 [35]). However, without proper communication methods, even with accurate sales predictions, the results may not be fully understood by the stakeholders. Dimara and Stasko [8] emphasized that visualization of the data can significantly enhance communication among decision-makers and other key stakeholders. As professional analysts are often responsible for conducting analysis and creating prediction models [12], it is crucial for them to have a comprehensive understanding of the business needs and strategically tailor their delivery methods to align with the specific requirements and expectations of stakeholders. Currently, prediction model development and deployment are often created by professional analysts, who act as a “middleman” between the data and decision-makers. This is because decision-makers are often unable to conduct an effective analysis. This can lead to distrust of the generated model use, as decision-makers lack visibility on data processing, analysis, and visualization process [12]. Perkhofier et al. [22] found that lack of knowledge and experience regarding the advanced analysis techniques is the main barrier of prediction practices. Therefore, utilizing a visualization dashboard as the deployment method for the prediction model serves as a flexible and easy-to-use tool for the industry to explore, analyze, and present the data [33]. This approach ensures that the insights provided are not only accurate and relevant but also actionable and effectively communicated, thereby facilitating better decision-making and alignment with the overall business strategy.

Despite the technological advancements that enable data-driven decisions and can bring significant opportunities for businesses, it is important to acknowledge the challenges faced by small-and-medium-sized enterprises (SMEs) in adopting and leveraging business intelligence, machine learning techniques, and data visualization [17]. Papachristodoulou et al. [21] also shared that BI is not primarily accessible to SMEs due to the difficulty of using the BI system and the requirement of technical knowledge and training. Recognizing these challenges is the first step towards finding solutions and providing the necessary support for SMEs to overcome them.

Currently, there are multiple frameworks that can be used to conduct data analytics projects, such as Business Analytics Life Cycle, Data Analytics Lifecycle,

Analytics Solutions Unified Method for Data Mining (ASUM-DM), and Cross Industry Standard Process for Data Mining (CRISP-DM). However, these frameworks are not well established [28], resulting in only 18% of the data science team following the framework explicitly [28], while businesses, more specifically SMEs, are still struggling to identify suitable practices for data analysis projects. This is due to the cost, lack of knowledge, and complexity of the tools [12]. This further highlights the importance of identifying suitable practices for industry to obtain and deploy an optimal prediction model. With data velocity and volume ever increasing, SMEs are struggling to develop suitable practices for analysis in the era of big data [4]. Even with the most commonly practiced methodology in the industry, CRISP-DM, providing a structured approach for business to conduct the analysis, it does not contain detailed elaboration of each step, techniques, and tools to follow, understand the procedure, and produce a complete solution [18]. As highlighted in the review by [28] on CRISP-DM, not all studies described how the data preparation was made, and the deployment phase were found excluded in most of the papers. While there were frameworks developed enhancing the CRISP-DM to meet the specific needs, such as APREP-DM (a framework for automating the preprocessing of a sensor data analysis) and LDTM (lean design thinking methodology for machine learning and modern data project), there is no significant number of studies using the enhancements [28].

Furthermore, over the years, various prediction models have been built to forecast sales/revenue. However, to the best of the researcher's knowledge, the settings used for prediction models for real business data and the reasons for using these settings were often not included in the discussion in prior research papers. Mitra et al. [19] developed 5 different prediction models for sales forecast of a retail chain, and Thivakaran and Ramesh [30] developed 6 different sales prediction models for a big mart. However, these research papers did not provide the settings used for model development and the reasoning behind these settings, underlining the need for more comprehensive and detailed research in this area. With numerous algorithms available for building prediction models, it is crucial to identify the most robust and accurate practices for revenue prediction in business settings. Lastly, as discussed in [28]'s research, the deployment stage was often omitted in the prior research. The focus of the previous research was to build and evaluate the models only, without providing guidelines to conduct deployment. The deployment is a crucial phase to transform the model from a theoretical tool to a practical component of business operations, ensuring the insights generated are used for making real-world data-driven decisions. With deployment, the model performance can also be continuously monitored and controlled [28].

This research aims to tackle a key challenge faced by SMEs: developing effective predictive models for forecasting revenue. The central question guiding this study is as follows: *What are the best methodology settings for developing predictive models for revenue forecasting for SMEs, and which machine learning model offers the most reliable performance for accurate revenue prediction?*

The study contributes to the field of business data mining by identifying an approach that works best for SMEs. These businesses often struggle to use advanced

machine learning techniques due to limited expertise and resources. By proposing a practical and easy-to-follow methodology, this research shows how SMEs can take advantage of data to improve their financial decision-making. This research fills a gap in existing studies by applying machine learning to the specific needs of SMEs, an area that has not been widely explored. It offers both theoretical insights and practical solutions, showing how machine learning can help improve revenue prediction and financial decision-making in smaller businesses.

2 Literature Review

Weytjens et al. [37] highlighted that predicting cash flow is essential due to its stochastic nature and limited managerial control. Numerous studies have sought to identify optimal prediction models across various fields, extending beyond traditional machine learning algorithms to explore hybrid approaches, which have often produced promising outcomes. For example, Haselbeck et al. [13] compared classical forecasting methods, such as exponential smoothing and seasonal ARIMA, with machine learning techniques including lasso regression, artificial neural networks (ANN), and extreme gradient boosting (XGBoost), finding gradient boosting to be the best performer. Similarly, Mitra et al. [19] also reported gradient boosting as a top-performing model but noted its susceptibility to overfitting, which can compromise reliability.

Random forest (RF), an ensemble learning method, is well-regarded for reducing model variance and improving predictive stability [20] and is widely used in feature selection, classification, and prediction tasks (Chen & Zhang, 2021). Hu and Szymczak [14] applied RF to predict the stock market using 16 indicators, demonstrating its ability to handle large datasets while maintaining high predictive accuracy. Moreover, Özen [20] further emphasized RF's capacity to capture diverse relationships between features and target variables, reducing inter-tree correlation and enhancing robustness against noise and outliers.

Hybrid models have attracted increasing attention for their ability to integrate the strengths of multiple algorithms while mitigating their individual limitations. Mitra et al. [19] proposed a hybrid RF–XGBoost–linear regression model for retail sales forecasting, outperforming single-model approaches such as RF, XGBoost, and ANN. They also stressed the importance of incorporating domain knowledge to enhance forecasting performance in different industries. Similarly, Udokwu et al. [34] demonstrated that combining a hybrid model with clustering techniques improved retail sales predictions and recommended developing modelling methods that optimize data preprocessing strategies.

ANNs have also been shown to be highly effective prediction tools. Studies on cash flow forecasting by Gupta and Nigam [11], Iskandar et al. [15], Son et al. [29], Weytjens et al. [37], and Hu and Szymczak [14] reported that ANN models often deliver superior accuracy compared to other algorithms. However, Thivakaran and Ramesh [30] have pointed out that achieving such performance requires rigorous and well-executed data preprocessing steps. Table 1 summarizes the most related studies.

Table 1 Summary of the related research

Model	Paper	Summary
XGBoost	(Haselbeck et al., 2022)	found Gradient Boosting to be the best-performing algorithm among several machine learning models. The researchers also found the influence of the data size on the model performance.
	(Thivakaran & Ramesh, 2022)	found decision tree-based algorithms, such as Gradient Boosting and Random Forest models, to be superior for small-to-medium structured data.
	(Mitra et al., 2022)	found Gradient Boosting to be one of the highest performing models for sales prediction. However, overfitting problem occurred.
Random Forest Regression	(Hu & Szymczak, 2023)	showed the random forest algorithm's ability to cope with big data, while generating high prediction accuracy.
	(Özen, 2024)	highlighted random forest algorithm's ability to capture different aspects of the relationship between the input features and target variable reduces the correlation between the trees, making it robust to noise or outliers in the data.
Hybrid Model	(Mitra et al., 2022)	developed a hybrid model, based on 'Random Forest – Gradient Boosting – Linear Regression, and found that it outperformed other machine learning models with single algorithm. It pointed out tailoring the model with adequate domain knowledge is crucial to improve the performance.
	(Udokwu et al., 2022)	A hybrid model using clustering algorithm was deployed to predict retail stores performance. 'K-Means – Random Forest' produced the best forecasting results. The researchers suggested to develop a modelling method that optimizes data preprocess for improved performance.
ANN	(Gupta & Nigam, 2020)	Researchers explained Artificial neural network (ANN) is based on the biological neural networks of the human brain, which are conceptually similar but not identical computational models.
	(Hu & Szymczak, 2023; Iskandar et al., 2018; Son et al., 2019; Weytjens et al., 2021)	These research showed ANN prediction models for cash flow prediction, have outperformed and produced high accuracy compared to other machine learning algorithms, such as logistic regression, random forest, support vector machine and so on.
	(Thivakaran & Ramesh, 2022)	found ANN to produce the highest performance to predict bigmart's sales. The researchers also noted that to produce successful prediction, the parameter settings and usage of other elements are important.

In addition, the industry needs to find a way to utilize machine learning and data analytics results for business users, as most of them have not acquired the necessary skills and capability to use machine learning and data analytics [38]. This can lead to distrust of the results, as business users lack visibility on data processing, analysis, and visualization processes [12]. Zimmermann and Brandtner [38] suggested dashboard creation to address this challenge. They highlighted that a dashboard can assist business users in exploring and understanding the data as well as insights generated by machine learning in a user-friendly and interactive manner.

The reviewed literature demonstrates significant progress in applying machine learning and hybrid models for forecasting tasks, yet several gaps remain. First, although gradient boosting and random forest have been identified as strong performers, their comparative effectiveness in real-world cash flow forecasting remains underexplored, especially in addressing overfitting and stability issues. Second, while hybrid models have shown promise, there is a lack of systematic evaluation of preprocessing strategies (e.g., outlier removal, feature engineering, or clustering integration) and their direct influence on forecasting accuracy. Third, despite evidence supporting the predictive power of ANNs, the heavy reliance on rigorous preprocessing poses challenges that are not consistently addressed in prior studies. Finally, an important practical gap exists in how model outputs are communicated to business users—current research provides limited solutions for translating advanced machine learning insights into accessible, trustworthy, and actionable decision-support tools, such as interactive dashboards.

By addressing these gaps, this study contributes to both the methodological development of predictive modelling and its practical implementation in business contexts.

3 Methodology

Cross Industry Standard Process for Data Mining (CRISP-DM) is one of the most widely used lifecycle frameworks for data science projects (Alogogianni & Virvou, 2021; [27]). [27] presented five strengths and five weaknesses of CRISP-DM for data science projects. The strengths shared are “Common Sense,” “Cyclical,” “Adoptable,” “Right Start,” and “Flexible.” Whereas weaknesses are defined as “Not a team coordination framework,” “can ignore stakeholders,” “outdated,” “documentation heavy,” and “slow start.” Despite the weaknesses, CRISP-DM is often used for various data science projects, as it encompasses a comprehensive process. Its structure allows flexibility and revisiting the previous phases, which aids in improving the outcome [7] and assists decision-makers by analyzing complex data to draw data-driven decisions [6].

In [6]’s research, CRISP-DM was utilized to evaluate the fintech industry’s risk via predictive, classification, and clustering analysis and successfully draw meaningful insights. However, the framework lacks a thorough explanation of procedures, methods, and tools to be used. As stated in [28]’s study, not all prior research which used CRISP-DM described how the data was prepared, and deployment was made. Although frameworks like APREP-DM (a framework for

automating the preprocessing of a sensor data analysis) and LDTM (lean design thinking methodology for machine learning and modern data project) were developed to enhance the CRISP-DM to meet the specific needs, not many studies have used the enhancements [28].

Before advancing to the creation of the machine learning model in a data mining project, developing the statistical prediction model can provide diagnostics on the model's fit with the data. Additionally, as demonstrated by Puttero et al. [24], the statistical model can be utilized as a benchmark for machine learning models. The linear regression model is a fundamental statistical model that is used for analyzing and modelling the relationship between a dependent variable and independent variables by fitting a linear equation to the data [24]. According to Puttero et al. [24] linear regression provides a simple yet powerful method to make predictions and understand the underlying relationships in the data. While Priya et al. [23] and Puttero et al. [24] developed different models based off the linear regression model, optimizing the performance of the statistical model itself was not discussed. Gerretzen et al. [10] highlighted that, while data preprocessing has a considerable influence on model performance, the absence of a systematic preprocessing selection approach can lead to a suboptimal preprocessing strategy. To improve the model performance, Gerretzen et al. [10] suggested a systematic approach based on design of experiment (DoE) to evaluate and determine the appropriate preprocessing methods for the given data. Based on the potential of DoE, this systematic approach inspires confidence in its effectiveness and reassures the researchers about its application in their research.

Based on the research objectives and prior research review, an enhanced CRISP-DM framework is developed, as shown in Fig. 1. This enhanced CRISP-DM framework is utilized to (1) develop the statistical prediction model with optimal setting and machine learning model to optimize the performance matrices using financial data and (2) develop a financial dashboard monitoring system that monitors the performance of the company's revenue, cost, and financial health. The enhanced CRISP-DM framework provides a guideline for the study to follow while obtaining quality results for both prediction and data visualization.

The enhanced CRISP-DM starts with the business understanding phase as the original CRISP-DM and moves back and forth with the data understanding phase. In data understanding, descriptive data analysis supports gaining an understanding of the main features of the dataset by utilizing statistical summary and/or visual analysis based on the data characteristics. Moving on to the data preparation phase, single-factor analysis is suggested to aid in assessing the impact of the factors and driving the optimal data preprocessing setting for model development while lowering the computational cost. Next, the "Modeling" phase has been divided into (1) statistical prediction model and (2) machine learning model. Starting the modelling by developing a statistical prediction model can offer diagnostics on the model's fit with the data and potential underlying data issues, such as multicollinearity, before moving to machine learning model development. Moreover, the statistical model establishes a baseline performance that the machine learning model is required to surpass. Then, machine learning models are developed to improve the error size and overall performance of the statistical

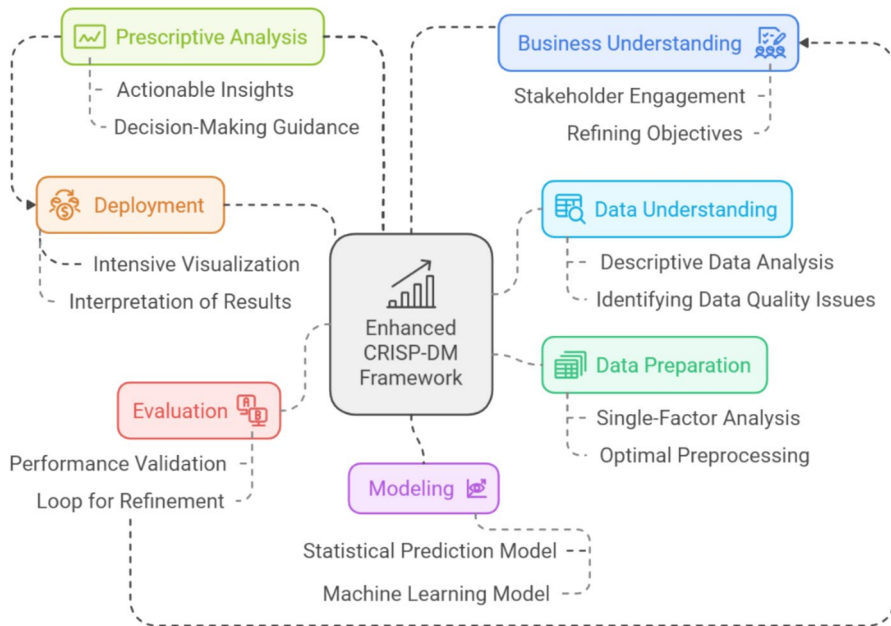


Fig. 1 Enhanced CRISP-DM framework

model. Predictive analysis is added as an alternating phase to develop the optimized machine learning model.

Once the machine learning model development is completed, an evaluation of the model is conducted to verify and validate the model's performance. At this stage, if the result does not meet the business objective, the project is circled back to the first phase and starts a new cycle. If the evaluation results are found successful, deployment is made. In this enhanced CRISP-DM framework, the deployment phase has been updated to "Deployment using visualization tools," as a visual representation of the data monitoring system would assist the stakeholders involved to understand the data while providing an accessible and interpretable format. A prescriptive analysis phase is added to ensure the visualization presents the actionable recommendations.

3.1 Data Collection

Real data from a small company was used for this study. The dataset, which was produced between July 2022 and June 2023, consists of a total of 13 columns and 1703 rows. While using "Invoice line amount" (revenue) as dependent variable, "cost of sales amount" (COGs), "TALA Category" and "Item Group" are chosen as independent variables for this research. Table 2 provides a summary of the data.

Table 2 Data structure

Column	Data Type	Data Description
A/R Invoice	String	Invoice Number
Date.Fiscal YQMD	Date	Fiscal date
G/L accounts	Integer	Account number
Sales unit	String	Unit of sales made
Currency	String	Currency of sales made
Item No.	String	Item number
Item groups.Item group	Nominal	Product item group
Delivery no.	String	Delivery number
TALA Category.TALA Category	Nominal	Product Category
Invoice line amount	Float	Revenue
Invoiced Qty	Float	Quantity of item invoiced
Delivered quantity	Float	Quantity of item delivered
Cost of sales amount	Float	COGs

Table 3 Data preparation

Data Preparation					
Data Cleaning	Remove any missing or duplicate data				
Outlier	Z-Score	IQR			
Train and Test Split Ratio	60:40	70:30	90:10		
Random_State	42	50	5	78	136
Cross Validation Folds	5 Folds	10 Folds			

3.2 Data Understanding

For data understanding, Power BI is utilized, as it is an interactive data visualization tool. Power BI acts as a platform not only to gain a comprehensive understanding of the data parameters without creating multiple charts/graphs but also to identify possible outliers or noise that may be presented in the data. A scatter plot is used to understand the relationship between the dependent variable and the independent variable. If categorical data is presented, a bar chart can be used to compare and understand the distribution in each category. Time trends and the relationship between the dependent variable and the independent variables can also be examined.

3.3 Data Preparation

The proposed data preparation is shown in Table 3. First, any missing or duplicate data that cannot be explained is removed. Then, four different factors are in consideration for the data preparation stage, which are (1) outlier treatment method, (2) train-test split ratio, (3) random_state value selection, and (4) cross-validation fold counts. Each of the factors' impact on prediction model performance is examined

through single factor analysis, and the most optimal and least optimal settings are determined.

Two methods are proposed for outlier treatment: Z-score and IQR. For the train-test split ratio, the three most used ratios in previous research were experimented with 60:40, 70:30, and 90:10. Meanwhile, for the `random_state` value, five different values were tested. Besides value “42,” which is the data science community’s standard practice `random_state` value, four other values were randomly selected. In this project, different `random_state` values were applied to generate varying train–test splits, which allowed us to examine how changes in data partitioning affected model performance. Lastly, two commonly used cross-validation fold counts, 5-folds and 10-folds, are tested.

3.4 Statistical Prediction Model

After data preparation is completed, to identify the optimal factor combination setting for prediction modelling, the design of the experiment (DoE) is conducted using JMP software. The DoE is conducted based on the results of the single factor analysis. “Fit Model - Standard Least Squares” and “Manova (Multivariate Analysis of Variance)” features are used to analyze the DoE result. The “Fit Model - Standard Least Squares” feature is designed to evaluate the relationship between variables and a linear model to the data by using the standard least squares method, which minimizes the sum of the squared differences between observed and predicted values. Whereas “Manova” is a statistical method used to analyze the difference in multivariate means across multiple groups and determine the impact of different factors on the outcome (*JMP Help*, n.d.-c). Based on the results of DoE, a statistical model is built, which can act as a baseline of machine learning model development. For the statistical model, a simple linear regression model is built using revenue as a predictive value (y) and COGs as an independent variable (x).

3.5 Development and Evaluation of Machine Learning Model

To improve the performance of the statistical model, three machine learning algorithms are chosen for the development of the revenue prediction model based on the literature reviews, which are (1) XGBoost, (2) random forest regression (RFR), and (3) artificial neural network (ANN). The data preparation for machine learning models follows the same optimal factor setting identified in DoE. In case categorical data is presented in the dataset, an additional data preparation step is taken, which is encoding the categorical values to numerical values. This is because the proposed algorithms cannot process categorical values. If the performance of the XGBoost and RFR model is lower than the statistical model performance, the development of a hybrid model is suggested. For hybrid model development, as proven to produce better performance in [19] and [34] combining the clustering algorithm with XGBoost and RFR is suggested. To ensure the optimization of the model performance, predictive analysis is conducted for each model built. XGBoost and RFR prediction models are built and trained. Hyperparameter tuning for model building

Table 4 Hyperparameters used for XGBoost, RFR, and ANN models

Model	Hyperparameter	Value
XGBoost	subsample	0.6
	n_estimators	100
	max_depth	3
	learning_rate	0.1
	colsample_bytree	0.8
Random Forest Regression (RFR)	max_depth	10
	min_samples_leaf	4
	min_samples_split	2
	n_estimators	200
Artificial Neural Network (ANN)	number of layers	3
	epochs	100
	batch_size	32
	learning_rate	0.001

is conducted using the GridSearchCV function from the sklearn.model_selection package. The GridSearchCV function systematically works through multiple combinations of parameter values, cross-validating as it goes to determine which combination of parameters gives the best model performance. The key hyperparameters used for all machine learning models are summarized in Table 4. These include the primary configuration settings for XGBoost, RFR, and the ANN, covering structural design, learning rate choices, training cycles, and other parameters that influence model performance.

If the single-algorithm machine learning models' performance is lower or insignificant improvement is made compared to a statistical model, a hybrid model is proposed.

ANN is used to develop a prediction model using Python's "TensorFlow" library. The process of building the prediction model is similar to XGBoost and RFR models. However, additional steps are added in building the ANN, which is DoE for the hyperparameter setting. During this DoE, the most optimal combination of activation and optimizer that can produce good ANN performance will be explored using JMP software. As model performance measurement, R^2 , MAE, MSE, and RMSE are used. Once the hyperparameter setting is determined, the ANN model is built with three layers, which consist of (1) the input layer, (2) the hidden layer, and (3) the output layer. During the ANN model training, callback functions from Keras, such as "EarlyStopping" and "ReduceLROnPlateau" are utilized. These callback functions assist in preventing overfitting by stopping the training when model performance is not improving and reducing the learning rate to optimize the training process.

Evaluation of the model performance is conducted using R^2 , MAE (mean absolute error), MSE (mean squared error), and RMSE (root mean squared error). These measurement metrics are popularly used to evaluate the performance of regression models. Each of the metrics provides different insights into the

model's performance. R^2 , which is the coefficient of determination, can provide insights into the model's goodness of fit, while MAE, MSE, and RMSE provide insights into prediction accuracy and error characteristics. These metrics are also utilized to compare the performance of the predictive models built and determine the highest-performing model.

3.6 Deployment Using Visualization Tools

For the deployment of the most optimal model, Power BI is employed to visualize the prediction outcome and develop a financial dashboard monitoring system that can provide insights regarding the company's financial health. During this stage, the actual value and the predicted value are examined by a scatter plot, which will provide insight into the model performance. Funnel graphs are utilized to explore the error value and the percentage difference between actual and predicted revenue. The dashboard also provides the current financial status, utilizing funnel graphs, scatter plots, and line graphs to present the revenue and cost trends and status. Table 5 summarizes the methodology used.

A pseudocode-style summary of the complete analytical pipeline is provided in the Appendix A. The pseudocode outlines the key stages of the study, including data preprocessing, single-model training, hybrid model construction, model evaluation, and deployment.

4 Results and Analysis

4.1 Data Understanding: Descriptive Analysis

Power BI was used to visualize the data and provide a comprehensive descriptive analysis. Figure 2 shows the overview of the sales made while showing the breakdown per "TALA Category (the company product category)" and "Item Groups." A funnel graph was utilized to present the percentage difference between revenue and COGs at a glance. The funnel graph shows that COGs account for 81.7% of the company revenue. To better understand the factors influencing the relationship between revenue and COGs in the company, further examination of the TALA category and Item Group was conducted using a "line and column chart." The figure also shows the relationship between TALA Category and Item Group and was conducted to gain a more comprehensive view of the data (Fig. 2b). In addition, observations of revenue and COGs are made regarding time trends and linear relationships (see Fig. 2c). Observation on the scatter plot for revenue and COGs shows that while the linear relationship is presented, there are outliers and a wide range of data points.

Table 5 Summary of methodology

Steps	Description	Expected Output
Data Collection	Real data from a small company was used.	Independent Variables: Cost of sales amount' (COGs), 'TALA Category' and 'Item Group'; Dependent Variable: 'Invoice line amount' (Revenue).
Data Understanding	Descriptive analysis using Power BI to explore the characteristics of the data.	1. The relationship between revenue and COGs can be examined using a scatter plot. 2. Bar chart to compare the values of different categories. 3. Line graph to understand the time series data.
Data Preparation	Removing any missing or duplicate data. Different data preprocessing settings, 1) outlier treatment method, 2) train-test split ratio, random_state value and cross-validation fold counts will be examined via single factor analysis.	Data is cleaned up, eliminating any missing or duplicate data. By conducting a single factor analysis on the data preprocessing setting, the most optimal and least optimal settings will be determined.
Statistical model development	A simple linear regression (SLR) model is developed, and predictive analysis is conducted. To develop the SLR model, different combinations of data preprocessing settings are examined by DoE using JMP software.	With DoE, the most optimal data preprocessing setting can be determined for statistical model development. Once the SLR model is developed, it can act as a foundation for machine learning model development.
Machine Learning model development and evaluation.	Several Machine learning models are used to develop a prediction model including 1) XGBoost 2) RFR and 3) ANN. In addition, hybrid models are developed by combining the clustering algorithm with XGBosot and RFR. The clustering algorithm suggested is 1) K-means and 2) K-prototype, depending on the data type. The model evaluation is conducted using a larger dataset. Performance metrics (R^2 , MAE, MSE, and RMSE) are compared between the models built with the original dataset and the larger dataset	Make improvements in the performance of the statistical model and optimize the model. Expected to produce a machine learning model that can have better model fit and lower error rate than statistical prediction model. The model performance is validated and verified to determine the most optimal model for revenue prediction.
Deployment using visualization tools	To deploy the prediction model, a financial dashboard monitoring system is developed using Power BI to visualize the prediction and the current financial health.	The prediction model's impact is visualized and provides insights into the influence of independent variables on prediction. Produce a monitoring dashboard, which the company can utilize to manage its financial health.

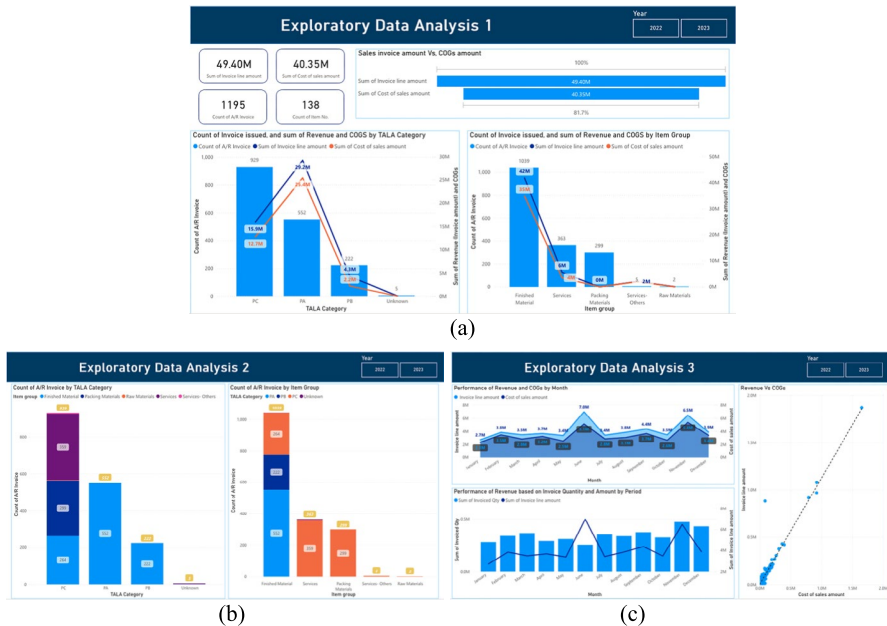


Fig. 2 Power BI exploratory data analysis

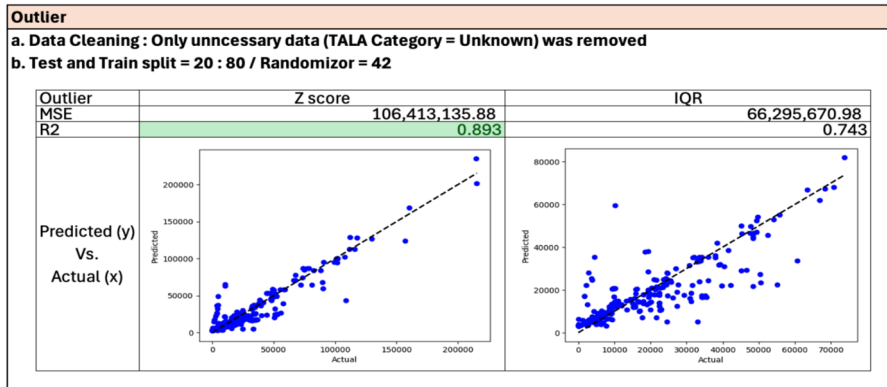


Fig. 3 Predicted vs. actual values on the test set after outlier treatment

4.2 Analysis of Data Preparation

After understanding the data, data preparation for the prediction model has been conducted. First, data that are with the TALA Category as “Unknown” were removed because the data point cannot be explained. Following the data cleaning, a single factor analysis was conducted to find the optimal data preparation setting for prediction modelling. First, the two outlier treatment methods, “Z-score” and “IQR,”

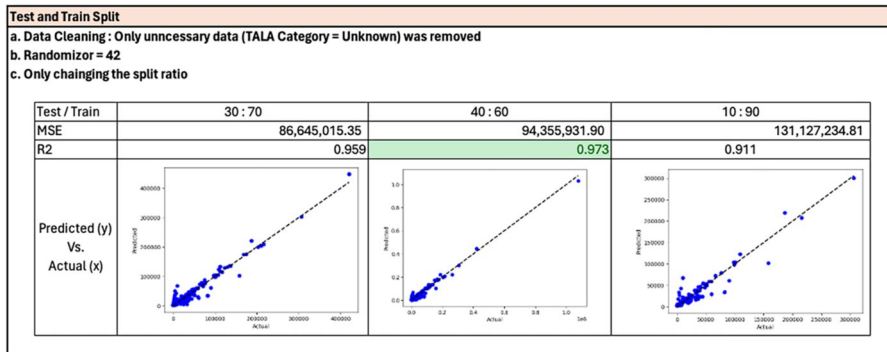


Fig. 4 Single-factor analysis—split ratio

were utilized with train-test split at 80:20 and random_state for “train-test” at 42. It was found while an SLR model with outlier treatment using “Z-score” shows higher MSE than “IQR,” the R^2 score for both models shows that the “Z-score” method is more suited to treating the outlier of the data (see Fig. 3). This also can be interpreted as the real data having a big range of spread, as the IQR method has shown to not result in good model fit.

Next, three different “Train-test split ratios” were tested to determine the impact of split ratios on the SLR model’s prediction. The experiment was conducted using the real data before the outlier was treated, with random_state at 42 for the “train-test” split. This experiment has shown that the split ratio has a significant impact on the performance of the SLR model, possibly caused by outliers and the range of the data (Fig. 4). Random_state for split was also experimented to determine its influence on the model performance. Two experiment conditions were set: condition (1) using data without outlier treatment with the train-test split ratio at 20:80, condition (2) using data with outlier treated using “Z score” with the train-test split ratio at 20:80.

Notably, using data without outlier treatment and a 20:80 train-test split, certain random_state values produced higher R^2 but also larger MSE, highlighting the sensitivity of results to data partitioning rather than to the random seed itself. It was found that setting random_state at “50” under condition 1 showed an R^2 score of 0.99, and its MSE score was showing relatively higher than another random_state setting under the same condition (see Fig. 5). This finding further highlights the impact of outliers present in the real data. Under condition 2, the overall MSE have lowered than condition 1, but so as R^2 . Table 6 summarizes the single-factor analysis results with different random-state values for two conditions, with and without treating the outliers.

Lastly, experiments were conducted to find suitable fold counts for cross-validation. The most widely used two counts of folds, 5 and 10, were used for the experiment. Figure 6 illustrates the results.

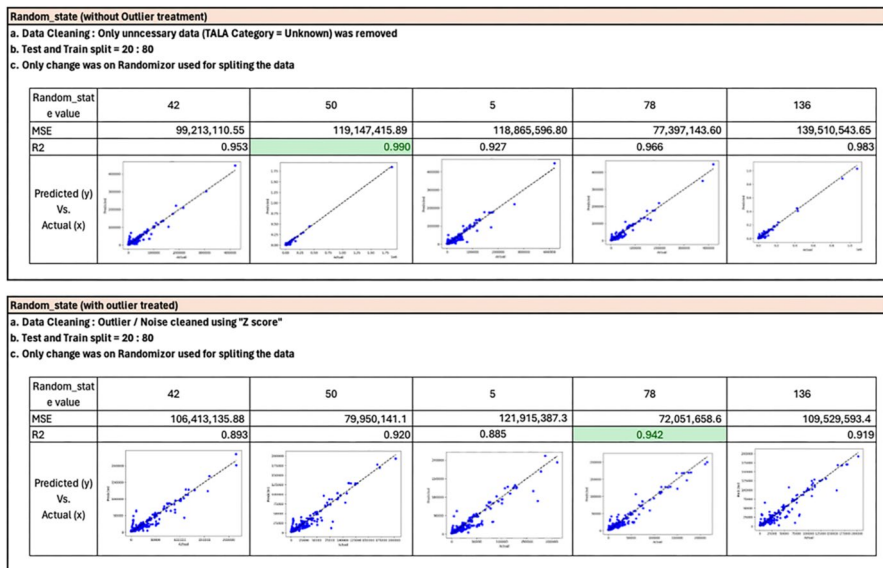


Fig. 5 Single factor analysis—random_state

4.3 Statistical Model Development

DoE was conducted using JMP to find the optimal factor combination setting for developing a statistical model, which is simple linear regression. DoE was set using the lowest and highest performance from each factor that was tested during data preparation (outlier treatment method, split ratio, random_state, and cross-validation fold counts). Table 7 shows performance measurement results with all settings combinations.

Using “Fit Model - Standard Least Squares,” effect screening was conducted. It was shown that “Outlier” has the most effect on the model performance, followed by “random_state and Outlier,” “random_state” and “split ratio and outlier.” Using prediction profiler, the most desirable factor combination was determined, which is Outlier with Z-score, random_state at “5,” split ratio at 10:90, and cross validation counts for 5 (see Fig. 7).

To further validate the factor combination, Manova was used to further analyze the impact. Manova results indicate that “Outlier” is the most influential factor, followed by “Randomize and Outlier,” “Random_state,” and “Split ratio and Outlier,” as illustrated in Fig. 8.

In addition, based on the DoE result, the parameter setting for SLR was set. The performance of SLR showed a good R^2 score (0.91), while MAE was at 6185, MSE was at 108,122,425, and RMSE was at 10,398 (see Fig. 9). These results indicate that the SLR model has a good fit with real data, but due to the dispersed range of the data, MSE/RMSE is large.

4.4 Machine Learning Model Development

To lower the error size in MAE/MSE/RMSE, different types of machine learning algorithms were used to build prediction models. The results showed that following the same data preprocessing setting as in statistical model development, the performance of the machine learning model was the worst when the random_state value is set as “5.” Therefore, based on the practice of literature, it was decided to change the random_state value to 42. Despite the promising results shown in other research conducted on regression-problem prediction, XGBoost’s performance on the real data showed similar results to SLR’s model performance, as illustrated in Table 8. In terms of the model fit, the XGBoost model showed 0.25% lower R^2 than the SLR model. However, for the XGBoost model’s error size, the overall MAE, MSE, and RMSE of the XGBoost model were lower than the SLR model. This result indicates that the XGBoost model’s fit with Company XY’s data is similar to the SLR model, while the wide range of MSE and RMSE indicates the presence of outliers in the data. Improvement in MAE, MSE, and RMSE of the XGBoost also indicates that the XGBoost model is more robust to outliers than SLR.

On the other hand, random forest regression (RFR)’s performance showed overall improvement over the SLR model. As shown in Table 9, while the R^2

Table 6 Single Factor Analysis – summary results

Condition 1	MSE	R2	Condition 2	MSE	R2
Average	110,826,762.10	0.964	Average	97,971,983.23	0.912
Max	139,510,543.65	0.990	Max	121,915,387.27	0.942
Min	77,397,143.60	0.927	Min	72,051,658.58	0.885
Standard deviation	23,500,169	0.025	Standard deviation	21,063,998	0.023

score is shown only 0.39% better than SLR, the RFR model had achieved lower MAE, MSE, and RMSE values compared to both the SLR and XGBoost models. The RFR algorithm demonstrates strong performance even in the presence of outliers, which accounts for its lower MAE, MSE, and RMSE values compared to SLR. The high MSE and RMSE of RFR also indicate the presence of extreme outliers or a high concentration of outliers in the data.

The difference in performance shown in Tables 8 and 9 can be easily explained. Random forest generates an ensemble of independent trees, which tends to be more robust than gradient boosting, which builds trees sequentially. For datasets with this kind of dynamic, random forest is often more favorable due to its stability and ability to generalize better in noisy conditions.

Cross validation fold counts						
a. Data Cleaning : Outlier / Noise cleaned using "Z score"						
b. Test and Train split : remained at 20% / 80%						
c. K-Folds Cross Validation / Scoring result is is Mean R2 score						
Random_ state value	42	50	5	78	136	Average
Fold = 5	0.917	0.910	0.921	0.906	0.909	0.913
Fold = 10	0.907	0.910	0.918	0.901	0.905	0.908

Fig. 6 Single-factor analysis—cross validation fold counts

Table 7 DoE for optimal factor combination setting

Rounds		Factors				Measure		
		random_state	split ratio	outlier	fold_counts	R2	RMSE	MAE
1ST	1	5	10 & 90	Z score	10 Folds	0.910608	9489.173	5382.927
	2	5	10 & 90	Z score	5 Folds	0.920179	9541.127	5379.538
	3	5	10 & 90	IQR	10 Folds	0.742151	8325.578	5105.484
	4	5	10 & 90	IQR	5 Folds	0.743795	8381.327	5097.821
	5	5	40 & 60	Z score	10 Folds	0.913437	9210.417	5417.075
	6	5	40 & 60	Z score	5 Folds	0.918982	9231.854	5417.194
	7	5	40 & 60	IQR	10 Folds	0.748214	8138.587	4993.098
	8	5	40 & 60	IQR	5 Folds	0.747919	8223.416	5005.208
	9	50	10 & 90	Z score	10 Folds	0.913944	9750.29	5642.026
	10	50	10 & 90	Z score	5 Folds	0.916276	9848.146	5652.878
	11	50	10 & 90	IQR	10 Folds	0.761032	8172.241	4990.207
	12	50	10 & 90	IQR	5 Folds	0.761645	8224.539	5002.432
	13	50	40 & 60	Z score	10 Folds	0.906755	9934.094	5892.35
	14	50	40 & 60	Z score	5 Folds	0.905376	10263.45	5892.312
	15	50	40 & 60	IQR	10 Folds	0.768166	7959.201	4708.974
	16	50	40 & 60	IQR	5 Folds	0.771269	8017.64	4715.041
2ND	1	5	10 & 90	Z score	10 Folds	0.910608	9489.173	5382.927
	2	5	10 & 90	Z score	5 Folds	0.920179	9541.127	5379.538
	3	5	10 & 90	IQR	10 Folds	0.742151	8325.578	5105.484
	4	5	10 & 90	IQR	5 Folds	0.743795	8381.327	5097.821
	5	5	40 & 60	Z score	10 Folds	0.913437	9210.417	5417.075
	6	5	40 & 60	Z score	5 Folds	0.918982	9231.854	5417.194
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	16	50	40 & 60	IQR	5 Folds	0.771269	8017.64	4715.041

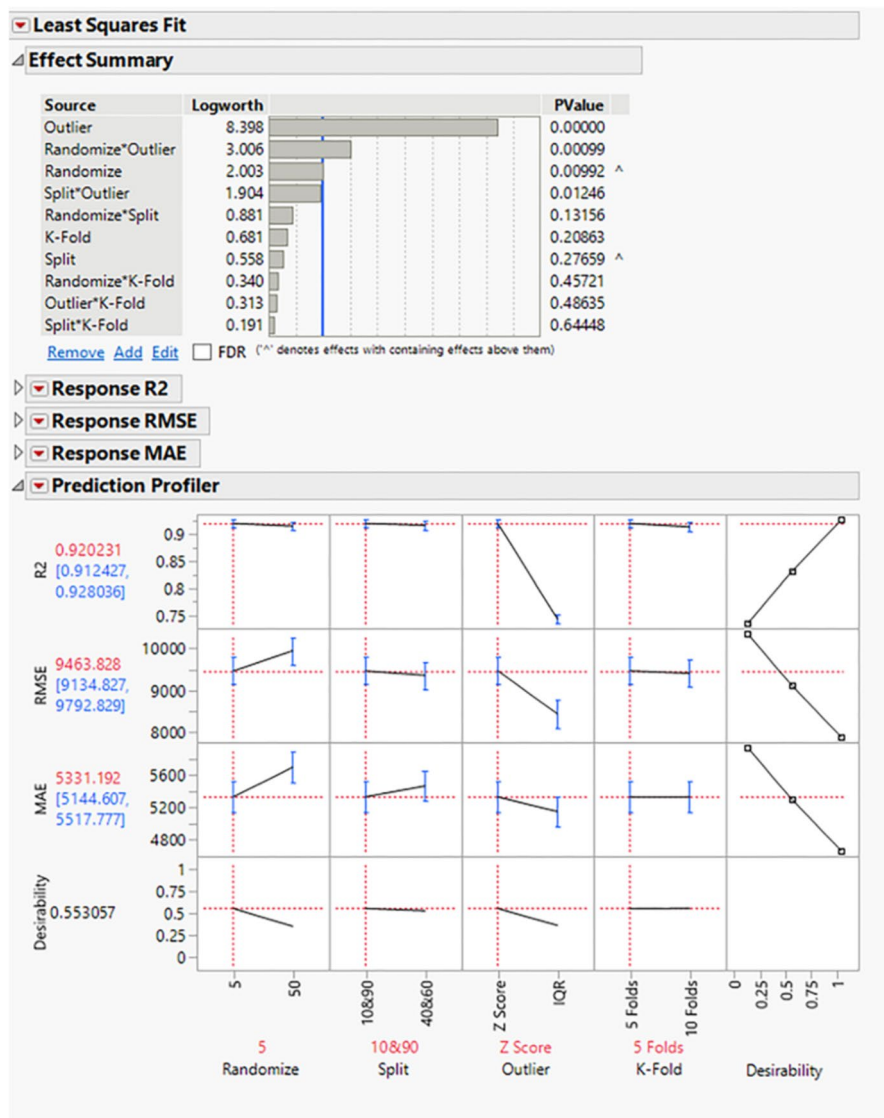


Fig. 7 DoE—effect screening

4.5 Machine Learning Model Optimization

While the prediction model using a single-algorithm lowers the error size on sales prediction, minimal improvement on the model fit was observed, but the error size is still considerably big. Therefore, hybrid model development has been conducted to further optimize the machine learning model. Large values of errors in terms of MSE and RMSE in XGBoost and RFR models indicate that the data has a

Parameter Estimates			
	R2	RMSE	MAE
Intercept	0.83435919	8919.44249	5268.41045
Randomize[5]	-0.0036986	-101.75763	-43.617214
Split[10&90]	-0.0006556	47.1100822	13.2537579
Outlier[Z Score]	0.07883535	739.126512	316.127211
K-Fold[5 Folds]	0.00132091	46.994794	1.89273712
Randomize[5]*Split[10&90]	-0.0008218	69.5061737	3.39557814
Randomize[5]*Outlier[Z Score]	0.00630538	-188.66856	-141.73676
Split[10&90]*Outlier[Z Score]	0.00271252	-48.494985	-83.449152
Randomize[5]*K-Fold[5 Folds]	0.00073722	-20.248719	-1.7455825
Split[10&90]*K-Fold[5 Folds]	0.00044911	-14.762674	-0.3894764
Outlier[Z Score]*K-Fold[5 Folds]	0.0006877	15.5806349	-0.9495515

Fig. 8 Manova—parameter estimate

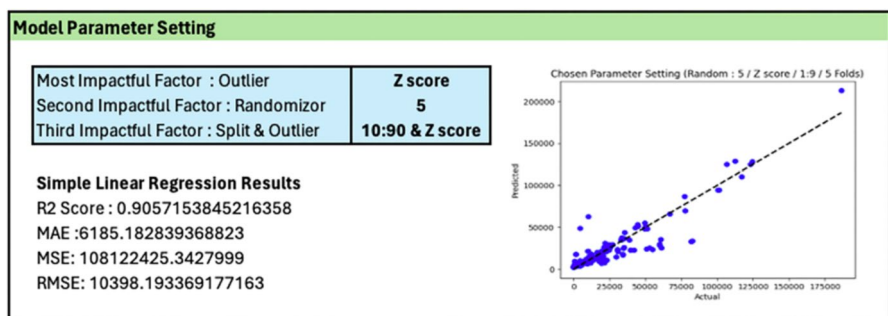


Fig. 9 Simple linear regression result

Table 8 XGBoost result

Model	R2	MAE	MSE	RMSE
Simple Linear Regression	90.57%	6,185	108,122,425	10,398
XGBoost	90.32%	5,552	87,242,690	9,340

Table 9 Random forest regression results

Model	R2	MAE	MSE	RMSE
Simple Linear Regression	90.57%	6,185	108,122,425	10,398
Random Forest Regression	90.96%	4,634	81,553,141	9,031

high concentration of outliers even after outlier was treated with Z-score methods. Accordingly, a clustering algorithm was utilized to understand the characteristics of the dimensional variables on the linear regression and its impact. The results of the

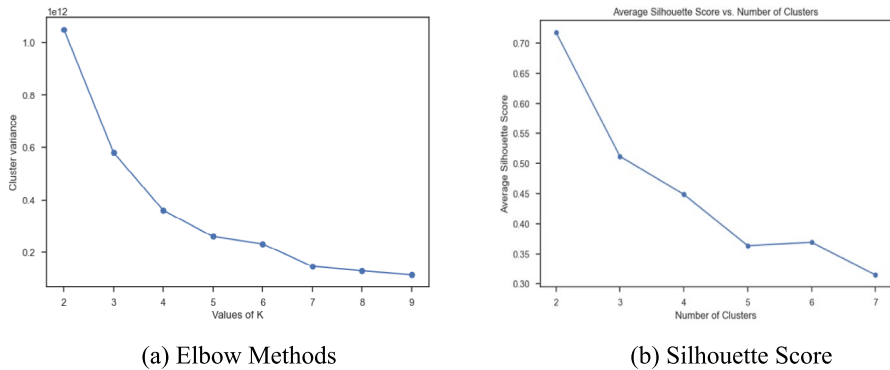


Fig. 10 Determining the optimal number of clusters

clustering algorithm enable the hybrid model better to capture the underlying relationships in the data and improve the predictive accuracy.

4.5.1 K-prototype

Due to the characteristics of the data, K-prototype was used to cluster the data. By examining “elbow methods” and “silhouette score,” the number of the clusters to group the data was 4. The choice of four clusters is supported by both the elbow method (see Fig. 10a), which shows a clear inflection at four clusters, and the silhouette plot (see Fig. 10b), where the average silhouette score is maximized at the same point, indicating good cluster separation and cohesion. To understand the clustered data, cluster analysis was conducted. When the clusters are visualized, the distribution of clusters in terms of revenue and COGs looked like in Fig. 11. While “cluster 1” shows the biggest spread among the clusters, outliers only are shown in “cluster 3” (see the box plot chart). However, upon examining the TALA Category and Item Group of these two clusters, the specific factors that might have caused such variation and outliers could not be identified. Therefore, further examination of the largest cluster 2 has been conducted.

Cluster 2 is the only cluster that contains all the TALA categories and Item Groups. That provided an opportunity to examine the TALA Category and Item Group’s impact on revenue and COGs within a smaller range of data. First, when looking at the scatter plot (Fig. 12) of revenue and COGs in cluster 2, significant amounts of data points are observed outside of the linear line.

To understand possible factors of outliers, box plot charts were generated for the TALA Category and Item Groups per revenue and COGs, as shown in Figs. 13 and 14. When examining the TALA Category, “PC” showed a significant number of outliers for both revenue and COGs, while “PA” and “PB” showed a bigger spread and skewness towards the third quartile in revenue than COGs.

By exploring Item groups, it is found that the “Services” groups have a significant amount of outliers, whereas “Packing Materials” shows no data, as the values

are measured at 0, and “Raw Materials” shows just a line due to a lack of data points available.

The analysis conducted on Cluster 2 provided possible factors that influence heavily on large values in MSE and RMSE, which are the “PC” TALA Category and “Service” Item Groups. A logistic regression model was used to validate the clustering results. The logistic regression model for predicting cluster labels showed an overall accuracy of 0.94 (Table 10).

4.5.2 Hybrid Model

For the first hybrid model, a combination of K-prototype and XGBoost was utilized. The overall performance significantly increased. In comparison to using XGBoost alone, not only did the R^2 score increase by 3%, but MAE, MSE, and RMSE showed dramatic improvement in a hybrid model (Table 11).

For the second hybrid model, the K-prototype and RFR combination were used. As with the first hybrid model, all performance measurement improvements were observed in the second hybrid model. The performance compared to when RFR was used alone is shown in Table 12 as follows: R^2 increased by 4.41%, MAE lowered by 1027, MSE lowered by 39,748,760 and RMSE lowered by 2565.

The performance improvement in both hybrid models indicates that clustering the data using the K-prototype has successfully captured the underlying relationships in the data and reduced the error range.

4.5.3 ANN Prediction

As the last machine learning model, the ANN model was built. Design of experiments (DoE) was conducted to identify the most optimal activation and optimizer combination via JMP tool. The ANN model was optimized using DoE to identify activation–optimizer combinations, while KerasTuner was employed to refine the number of layers and nodes. Additional callbacks (e.g., *EarlyStopping*, *ReduceLROnPlateau*, and ridge (L2) regularization) were incorporated to improve generalization and reduce overfitting. As shown in Table 12, activation: Relu and optimizer: Nadam combination showed the highest performance. This result was further verified by JMP software’s prediction profiler and Manova results, as shown in Fig. 15 and Table 13.

While ANN models were shown to have promising prediction performance in previous research, on the data, the performance was dismal. Model fit was 87.99% while showing higher error rates than other hybrid models, as shown in Table 14. One of the possible reasons for the poor performance of ANN could be due to the lack of data points in the dataset, as deep learning models perform the best when larger data points are presented.

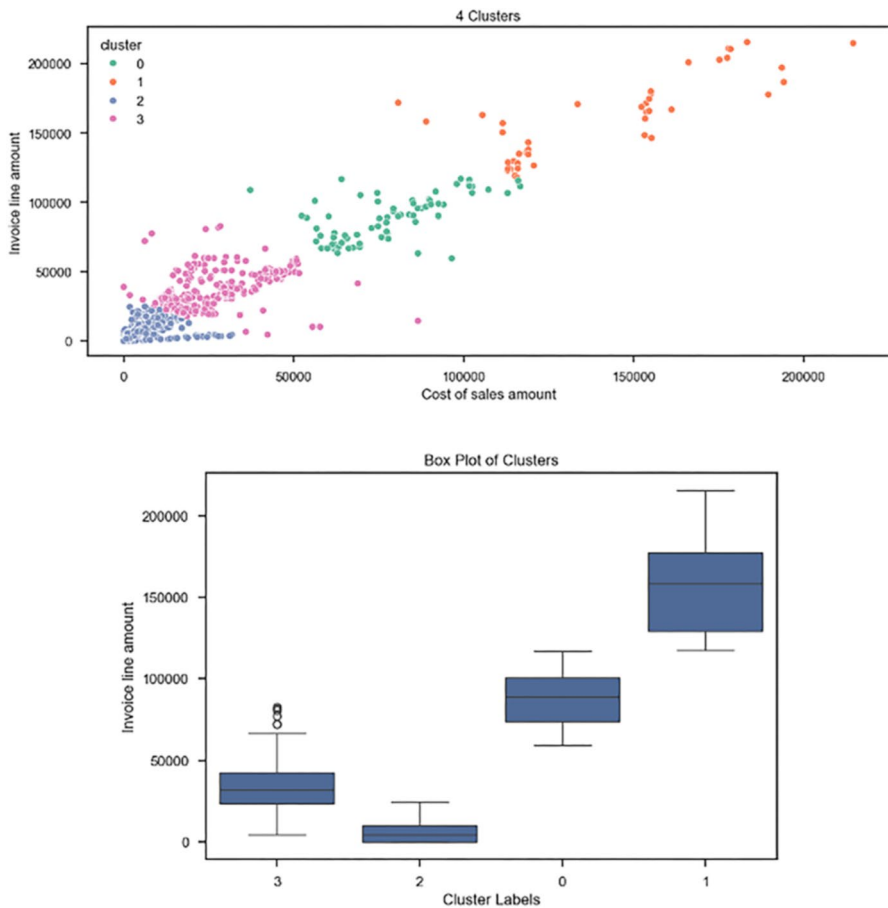


Fig. 11 K-prototype cluster visualization

4.5.4 Machine Learning Model Evaluation and Comparison

To validate the model performance with unseen data and reliability in the real-world scenario, data produced from July 2021 to March 2024, which consists of 4,329 rows, was used. Overall model fits and MAE score have shown similar performance in comparison to model performance with the original dataset. However, MSE and RMSE increased when validation data was used on the models. This indicates validation data has a wider range of data points than the original dataset. When validation data is used, hybrid models showed better performance than single-algorithm-used models, while the ANN model's performance was lower than single-algorithm-used models, as shown in Table 15. These results are aligned with the model performance of the original dataset. Additionally, despite the model overfitting observed when validation data was used for single-algorithm models (simple linear regression, gradient boosting, and random forest regression), hybrid models showed no sign of model overfitting.

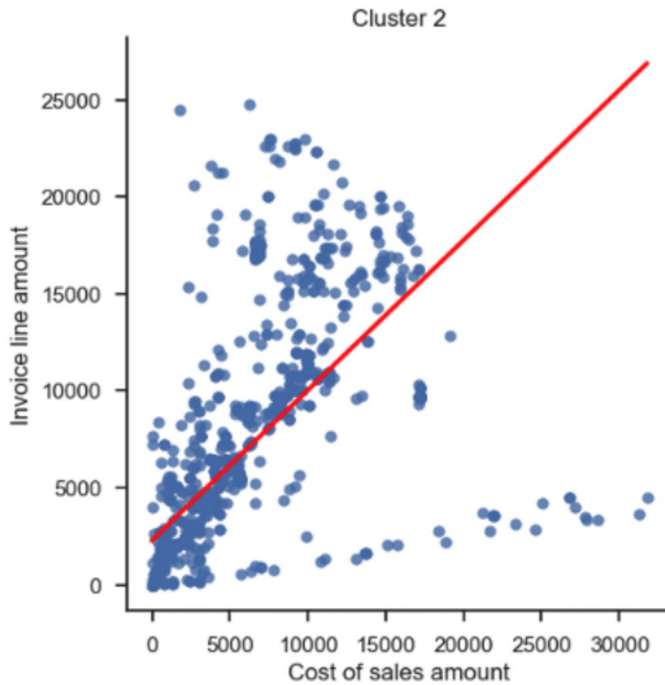


Fig. 12 K-prototype cluster 2—scatter plot

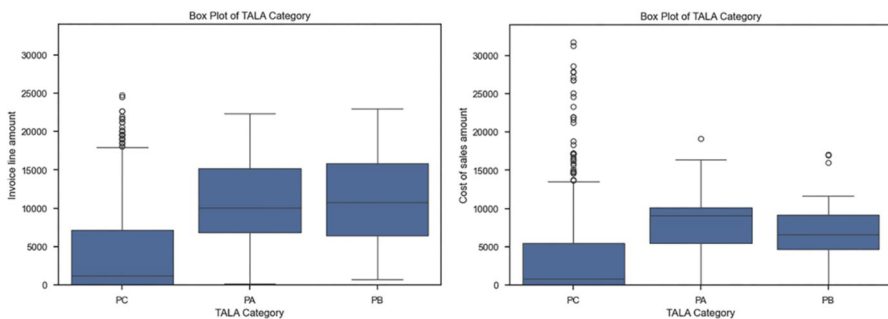


Fig. 13 K-prototype cluster 2—box plot for “TALA Category”

Furthermore, while the ANN model showed the lowest performance when used for real data, to validate the theory that the ANN model is not performing well due to the size of the data, the secondary dataset was fed to the ANN model built. The secondary contained 792,277 data points. The ANN model was used to predict “Revenue (y)” using “Original Selling Price” and “Unit Sold” as “x.” The ANN model’s fit with the secondary dataset was 96%, while it showed relatively

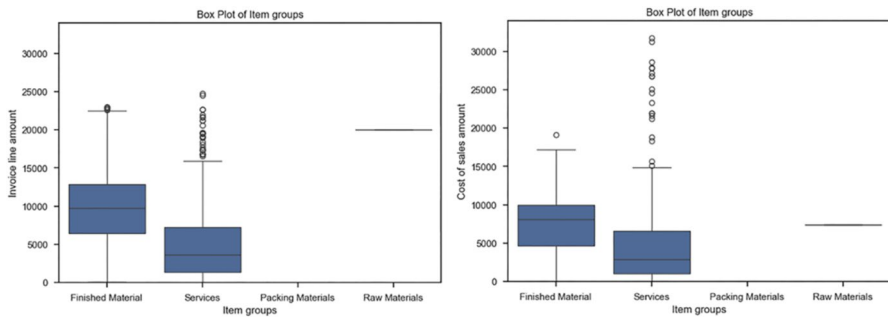


Fig. 14 K-prototype cluster 2—box plot for “Item Group”

smaller error scores (see Fig. 16). This experiment demonstrates the importance of data size when using deep mining algorithms.

Among the models tested in this project, both the hybrid models demonstrated comparable performance metrics (Table 16). However, during the validation process, it was determined the hybrid model using “K-prototype and random forest regression” showed the most consistency and optimal performance. Due to RFR algorithms’ strength in handling outliers, when combined with a clustering algorithm, the hybrid model has shown good model fit (94.03%) with the lowest MAE, MSE, and RMSE score.

To further evaluate the practicality of the proposed models, we examined their computational runtimes under the specified computing environment. As expected, hybrid models required longer runtimes than single-algorithm models due to the additional clustering step. The results are summarized in Table 17, which reports the average runtime for each model on both the original and validation datasets. These findings provide useful insights for practitioners, particularly those operating in resource-constrained environments, by highlighting the trade-offs between predictive performance and computational efficiency.

4.6 Financial Dashboard Monitoring System

A dashboard monitoring system was built using Power BI to deploy the prediction model. The results of the most optimal model, “K-prototype+random forest regression,” were used in the dashboard. The dashboard presents the prediction difference from actual data in the form of an exact amount (140.42 K) and percentage (−1.24%). It also shows how many test data points were used (433). The dashboard includes slicers for “Item Group,” “TALA Category,” and “Cluster,” which is the clustering label by K-prototype, to show the prediction impact based on these factors. A funnel graph visualized the percentage difference between actual revenue and predicted revenue to support the cognitive process. Multiple scatter plots visualized the relation between (1) actual revenue and COGs, (2) predicted revenue and COGs, and (3) actual revenue vs. predicted revenue. Figure 17 shows the dashboard for invoice prediction results.

Table 10 Logistic regression used for clustering validation

Cluster	Precision	Recall	F1-score	Support
0	0.89	0.80	0.84	10
1	0.75	0.75	0.75	4
2	0.99	0.94	0.96	97
3	0.89	0.98	0.93	57
Accuracy			0.94	168
Macro Avg	0.88	0.87	0.87	168
Weighted Avg	0.94	0.94	0.94	168

Table 11 K-Prototype and XGBoost hybrid model performance

Model	R2	MAE	MSE	RMSE
Simple Linear Regression	90.57%	6,185	108,122,425	10,398
XGBoost	90.32%	5,552	87,242,690	9,340
K-prototype + XGBoost	93.61%	3,397	68,744,448	8,291

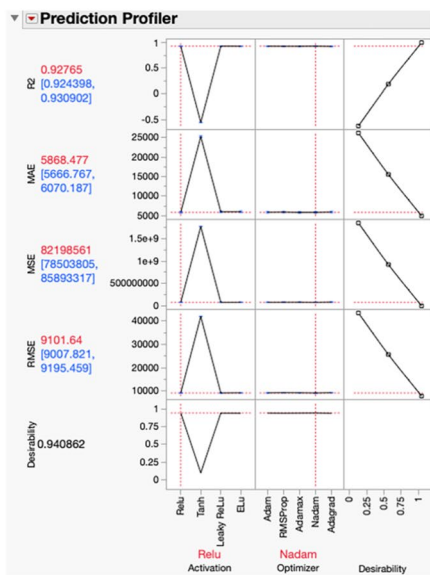
Table 12 K-Prototype and Random Forest Regression hybrid model performance

Model	R2	MAE	MSE	RMSE
Simple Linear Regression	90.57%	6,185	108,122,425	10,398
Random Forest Regression	90.96%	4,634	81,553,141	9,031
K-prototype + Random Forest Regression	95.36%	3,607	41,804,381	6,466

When examining the dashboard with different filter settings on the cluster label, it was shown that cluster 0 has a higher prediction error rate than other cluster groups. Cluster 0 is the only group that has TALA Category: “PA” and “PC,” and Item Group: “Finished Material” and “Services.” When choosing only “PC” in “Cluster 0,” the prediction showed a different percentage of -13.36% . This suggests that Company XY should focus on TALA Category “PC” products to further understand why such differences are showing. As demonstrated, the dashboard can assist the user, who may not have expertise in data analysis or prediction analysis, in understanding the impact of each cluster, TALA Category, and Item Group on the revenue prediction and in driving informed decisions in an interactive and user-friendly manner.

In addition, two interactive dashboards were built to present the status of financial health and monitor its revenue and COGs based on the products. The “Over the years invoice trends” dashboard (see Fig. 18a) is designed to provide an overview of the sales and COGs over the years, along with the main products sold in the company. The dashboard also contains revenue made per “Item Group” and “TALA Category.” On the other hand, the “Invoice and COGs” dashboard, as illustrated in Fig. 18b, provides a further analytical view of the invoice and COGs’ relationship. It also presents the product’s COGs sum, revenue amount, and actual invoiced quantity. Utilizing these dashboards, the dashboard can provide an opportunity for the company to analyze the main products sold and each Item Group/TALA Category’s product revenue and COGs’ trends and the relationship.

No	Activation	Optimizer	R2	MAE	MSE	RMSE
1	Tanh	Nadam	-55.83%	25,266	1,770,410,120	42,076
2	Tanh	RMSProp	-55.82%	25,266	1,770,371,719	42,076
3	Tanh	Adagrad	-56.86%	25,422	1,782,126,536	42,215
4	Tanh	Adam	-55.83%	25,266	1,770,390,712	42,076
5	Tanh	Adamax	-56.23%	25,326	1,775,027,001	42,131
6	Relu	Adagrad	92.46%	6,008	85,694,292	9,257
7	Relu	Nadam	92.71%	5,729	82,770,793	9,098
8	Relu	Adamax	92.50%	5,989	85,246,560	9,233
9	Relu	RMSProp	92.61%	5,984	83,944,090	9,162
10	Relu	Adam	92.64%	5,809	83,573,366	9,142
11	Leaky ReLu	Adam	92.70%	6,009	82,982,576	9,109
12	Leaky ReLu	Adagrad	92.58%	6,004	84,334,500	9,183
13	Leaky ReLu	Adamax	92.64%	5,849	83,664,199	9,147
14	Leaky ReLu	RMSProp	92.27%	6,325	87,854,020	9,373
15	Leaky ReLu	Nadam	92.68%	5,996	83,131,131	9,118
16	ELu	RMSProp	92.39%	5,882	86,472,934	9,299
17	ELu	Adagrad	92.44%	6,048	85,878,281	9,267
18	ELu	Adam	92.38%	6,213	86,576,028	9,305
19	ELu	Nadam	92.66%	6,163	83,349,355	9,130
20	ELu	Adamax	92.48%	5,923	85,453,962	9,244



Manova

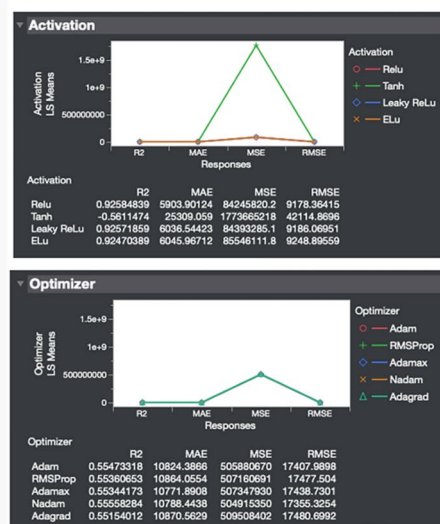


Fig. 15 ANN DoE results via Prediction profiler and Manova

5 Discussion

This research project aimed to identify the optimal methodology settings for a revenue prediction model using real data, the most robust optimal machine learning model for predicting revenue in an industrial environment, and to develop the optimal financial dashboard monitoring system that can assist stakeholders in monitoring the performance of the company's revenue, cost, and financial health.

Firstly, using real data, the most optimal practices for obtaining prediction model settings for data are obtained. To understand the data, descriptive analysis is conducted using EDA. During this phase, potential outliers and noise were found in the data. While a linear relationship between revenue and COGs was observed, time

Table 13 ANN DoE for hyperparameter setting

No	Activation	Optimizer	R2	MAE	MSE	RMSE
1	Tanh	Nadam	-55.83%	25,266	1,770,410,120	42,076
2	Tanh	RMSProp	-55.82%	25,266	1,770,371,719	42,076
3	Tanh	Adagrad	-56.86%	25,422	1,782,126,536	42,215
4	Tanh	Adam	-55.83%	25,266	1,770,390,712	42,076
5	Tanh	Adamax	-56.23%	25,326	1,775,027,001	42,131
6	Relu	Adagrad	92.46%	6,008	85,694,292	9,257
7	Relu	Nadam	92.71%	5,729	82,770,793	9,098
8	Relu	Adamax	92.50%	5,989	85,246,560	9,233
9	Relu	RMSProp	92.61%	5,984	83,944,090	9,162
10	Relu	Adam	92.64%	5,809	83,573,366	9,142
11	Leaky ReLu	Adam	92.70%	6,009	82,982,576	9,109
12	Leaky ReLu	Adagrad	92.58%	6,004	84,334,500	9,183
13	Leaky ReLu	Adamax	92.64%	5,849	83,664,199	9,147
14	Leaky ReLu	RMSProp	92.27%	6,325	87,854,020	9,373
15	Leaky ReLu	Nadam	92.68%	5,996	83,131,131	9,118
16	ELu	RMSProp	92.39%	5,882	86,472,934	9,299
17	ELu	Adagrad	92.44%	6,048	85,878,281	9,267
18	ELu	Adam	92.38%	6,213	86,576,028	9,305
19	ELu	Nadam	92.66%	6,163	83,349,355	9,130
20	ELu	Adamax	92.48%	5,923	85,453,962	9,244

Table 14 Hybrid models and ANN performance comparison

Model		R2	MAE	MSE	RMSE
Hybrid Model	K-prototype + XGBoost	93.61%	3,397	68,744,448	8,291
	K-prototype +Random Forest Regression	95.36%	3,607	41,804,381	6,466
ANN		87.99%	5,617	108,267,617	10,405

Table 15 Model performance evaluation with validation data

Model		Original	Validation	Original	Validation	Original	Validation	Original	Validation
		R2	R2	MAE	MAE	MSE	MSE	RMSE	RMSE
Single Model	Simple Linear Regression	90.57%	85.33%	6,185	7,198	108,122,425	239,871,036	10,398	15,488
	XGBoost	90.32%	86.09%	5,552	6,090	87,242,690	227,390,826	9,340	15,079
	Random Forest Regression	90.96%	86.36%	4,634	5,712	81,553,141	222,936,635	9,031	14,931
Hybrid Model	K-prototype + XGBoost	93.61%	92.97%	3,397	4,777	68,744,448	114,894,093	8,291	10,719
	K-prototype + Random Forest Regression	95.36%	94.03%	3,607	4,038	41,804,381	97,660,831	6,466	9,882
	ANN	87.99%	85.13%	5,617	6,327	108,267,617	243,118,851	10,405	15,592

trends related to the company's financial activities were also observed. During the data preparation stage, any missing and duplicate data were first removed, and single factor analysis was conducted to examine the impact of different data preprocessing steps: (1) outlier treatment, (2) train-test split ratio, (3) random_state, and (4) cross-validation folds count. Based on the single factor analysis results, DoE was conducted to determine the most optimal setting for statistical model development. The most optimal setting for the statistical prediction model was determined by DoE and conducted using JMP software. The setting for the statistical model was set as an outlier treatment with "Z-score," random_state at "5," train-test split ratio as "10:90," and cross-validation fold counts at "5." This setting resulted in the statistical model with an R^2 score of 0.91. Conducting DoE on data preprocessing settings

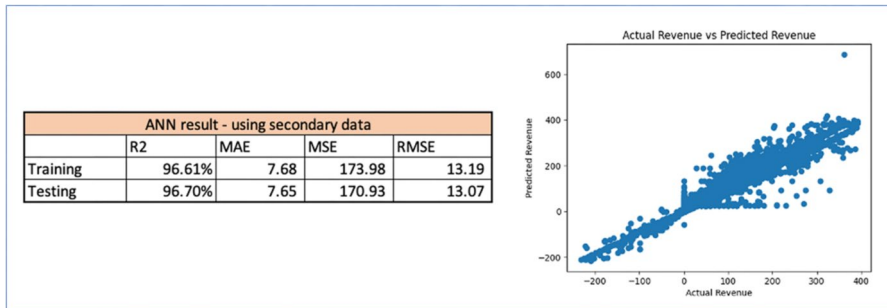


Fig. 16 Secondary data used for ANN model validation

Table 16 Prediction model comparison using validation data

Model		R2	MAE	MSE	RMSE
Single Model	Simple Linear Regression	85.33%	7,198	239,871,036	15,488
	XGBoost	86.09%	6,090	227,390,826	15,079
	Random Forest Regression	86.36%	5,712	222,936,635	14,931
Hybrid Model	K-prototype + XGBoost	92.97%	4,777	114,894,093	10,719
	K-prototype + Random Forest Regression	94.03%	4,038	97,660,831	9,882
ANN		85.13%	6,327	243,118,851	15,592

ensured the identification of the optimal combination of factors that can maximize data quality and improve model performance efficiently and effectively.

Secondly, the most robust revenue prediction machine learning model for the data is concluded as “K-prototype + RFR.” To ensure that the model is optimized, predictive analysis was conducted on every model built in this research, and the results indicate both hybrid models, “K-prototype + XGBoost” and “K-prototype + RFR,” outperform single-algorithm models and ANN, especially if a limited amount of data is available and the dataset’s range is wide with a high concentration of outliers.

While the comparative results highlight the strengths of ensemble and hybrid models, it is also important to reflect on the relatively weaker performance of the ANN. This underperformance can be better understood by considering ANN-specific challenges. These include their sensitivity to data distribution, difficulties in parameter tuning when training on small samples, and the risk of overfitting. To investigate whether dataset size was a limiting factor, we tested the ANN on a secondary dataset containing 792,277 records. With this larger dataset, the ANN achieved a substantially higher fit of 96%, suggesting that its performance is indeed sensitive to dataset size. Our findings align with prior research that highlights the influence of sample size and data balance on ANN performance. For example, Benkendorf and Hawkins [3] found that increasing sample size from 100 to 10,000 improved ANN performance nearly linearly, with deeper architectures prone to overfitting on small datasets. Similarly, Bailly et al. (2022) reported higher accuracy with larger datasets (up to 100,000 records), though they noted no consistent *significant* effect of dataset size alone. More recently, Jeong and Kim [16] showed that the minimum dataset size required for ANN modelling is closely tied to data balance, with imbalanced datasets requiring substantially more data to achieve reliable predictions. Based on both our

Table 17 Computational runtime of prediction models on original and validation datasets

Model		Computational Runtime	
		Original Data	Validation Data
Simple Linear Regression		1.62 seconds	1.81 seconds
Single Model	XGBoost	5.07 seconds	6.26 seconds
	Random Forest Regression	27.24 seconds	50.02 seconds
Hybrid Model	K-prototype + XGBoost	14.40 seconds	25.54 seconds
	K-prototype + Random Forest Regression	31.40 seconds	77.93 seconds
ANN		12.00 seconds	5.13 seconds

results and corroborating evidence in the literature, we recommend that practitioners use at least 1000 records with a balance ratio greater than 0.3 for ANN-based predictive modelling. More broadly, the methodology demonstrated in this research—combining systematic hyperparameter tuning with dataset-specific validation—can guide both practitioners and researchers in selecting the most appropriate machine learning models for different predictive tasks.

In comparing the performance of random forest and gradient boosting, the results in Tables 7 and 8 indicate that random forest achieved more stable and reliable outcomes. This can be explained by the fact that random forest generates an ensemble of independent trees, which collectively reduces variance and enhances robustness against noise in the data. By contrast, gradient boosting builds trees sequentially, where each new tree attempts to correct the errors of the previous ones. While this sequential learning often improves predictive power, it can also make the model more sensitive to noise and overfitting, particularly in datasets with complex dynamics or outliers. Therefore, the superior performance of random forest in this study is consistent with its reputation for stability and better generalization under noisy conditions. In addition, single-algorithm models (XGBoost and RFR) have shown an overfitting issue. Ramampandra et al. [25] shared that overfitting of the model occurs when the model exhibits “too much flexibility in the response shapes” or when the data provided is limited and exhibits high variability, which often manifests as noise that cannot be explained by the influencing factors. The overfitting issue shown for single-algorithm models (XGBoost and RFR) when validation data was used demonstrated that the overall data contains significant noise. However, the hybrid models built showed resilience against overfitting. While the “K-prototype + XGBoost” model showed lower performance with validation data, the “K-prototype + RFR” model maintained consistent performance. This indicates the strength of the RFR algorithm in handling outliers, and noise has an amplifying effect when combined with a clustering algorithm. While the hybrid model developed is different, this result aligns with the research conducted by [34] and [19], in which the hybrid model performance was comparatively better than the single-algorithm and ANN model as well.

Thirdly, this research project developed a financial dashboard monitoring system for deployment of the most optimal prediction model, “K-prototype + RFR,” using

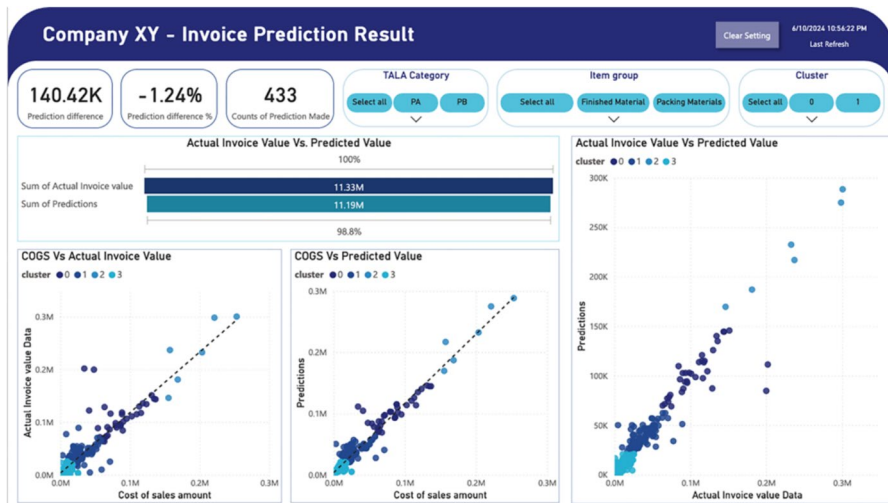
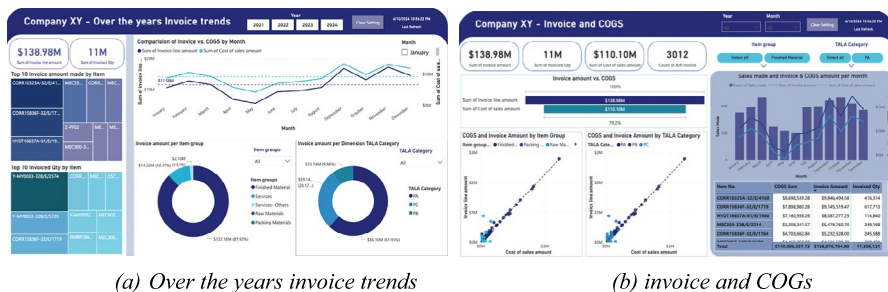


Fig. 17 Dashboard—prediction results



(a) Over the years invoice trends

(b) invoice and COGS

Fig. 18 Dashboard

the Power BI tool. The interactive dashboard included (1) prediction results and (2) the status of the company's financial health using various charts and slicers. Power BI showed a powerful ability to build an interactive dashboard that can assist stakeholders without background knowledge of data analysis in understanding the impact of the prediction and model analysis. This dashboard provides an opportunity to monitor and obtain insights on the impact of the company revenue and COGs based on the products, with easy-to-understand visualizations.

6 Study Implications

This study offers several theoretical implications. It reinforces prior findings on the strengths and weaknesses of widely used predictive models such as random forest, gradient boosting, and artificial neural networks, while extending the discussion on

hybrid approaches that integrate multiple algorithms. The results provide empirical evidence that hybrid models can outperform single-model methods, thereby supporting ongoing theoretical arguments for integrated forecasting frameworks. Additionally, the study highlights the critical role of rigorous data preprocessing, contributing to the literature by demonstrating its direct influence on predictive accuracy and overall model reliability.

From a managerial perspective, the findings offer practical guidance for organizations seeking to adopt machine learning for cash flow and retail forecasting. Managers can select suitable models based on their operational needs—for example, using random forest for stable and interpretable outcomes, or hybrid models when higher predictive accuracy is required. The study also emphasizes the importance of developing user-friendly dashboards and visualization tools to translate technical model outputs into actionable insights, thereby supporting more confident decision-making. Furthermore, the results underscore the necessity for organizations to invest in data quality and preprocessing capabilities, as these are foundational to achieving reliable forecasting outcomes that enhance financial planning and operational efficiency.

7 Limitations and Future Research

This study has several limitations that point to important directions for future research. First, a small number of records with missing TALA values were removed during preprocessing. While removing these records did not noticeably affect the results in this dataset, future studies should examine alternative ways of handling missing values such as model-based estimation to determine whether different approaches influence model stability and predictive accuracy.

Second, while advanced optimization and regularization techniques such as dropout, batch normalization, or more extensive hyperparameter searches could potentially enhance generalization, these methods were not fully implemented due to the relatively limited dataset size, which restricted their effectiveness. Future research using larger or more complex datasets could systematically evaluate these techniques to determine their impact on improving model generalization, particularly for ANN and hybrid architectures.

Third, although this study employed common evaluation metrics such as R^2 , MAE, MSE, and RMSE, no statistical significance testing was conducted. Incorporating tests such as bootstrap confidence intervals, paired t -tests, or non-parametric alternatives would enable researchers to determine whether observed differences in performance are statistically meaningful, thereby strengthening the evidential basis for comparing models.

Finally, future investigations could extend this work by exploring additional hybrid model configurations, applying explainable AI techniques to improve transparency for business users, and assessing deployment strategies across a wider range of SME environments.

8 Conclusion

The findings of this study contribute to both industry and academia by addressing critical gaps in the existing literature and offering practical applications. The project investigated the optimal methodology to obtain the revenue prediction model setting using real data by enhancing the CRISP-DM framework and explored various machine learning models to identify the most robust model to predict revenue with real data. Additionally, a financial dashboard monitoring system was also developed as a recommended deployment method of the prediction model, allowing the stakeholders to monitor their revenue performance and financial health in a user-friendly manner. These findings offer valuable insights and guidelines for the practical application in the industry for revenue prediction problems and developing financial monitoring systems while laying the foundation for future research and development for further improvements.

As the company immigrated to a new ERP system in 2021, not enough data was available. The small data size led the researcher to look at the analysis from various angles to minimize the error rate without damaging the integrity of the data. Future research, which follows the methodology that was built in this study, should consider utilizing a bigger data size. In addition, it is suggested that different types of machine learning models can be explored. As the robust performance of hybrid models was shown in this project, it is worth exploring other combinations of algorithms to further minimize the error size. Moreover, granted that the data size is big, it is suggested to explore other deep learning models, such as recurrent neural networks, long short-term memory networks, convolutional neural networks, and hugging face, which are popularly used in other research papers. Lastly, including more independent variables in prediction model development is suggested in hopes of reducing the error size even further. For manufacturing companies, it would be beneficial to consider the material component of products as one of the independent variables for revenue prediction. Examining material components will provide a further breakdown of COGs, which will add a layer for the prediction model to work with and provide an opportunity to improve the material management system.

Appendix A

Input: Raw transactional dataset **D**

Output: Trained predictive models and evaluated performance metrics

Start

Data Exploration

Load raw dataset **D**

Inspect data types, missing values, and outliers

Identify numerical and categorical features

Define target variable (**Revenue**)

Data Preparation

Remove samples with *unknown* or *invalid* target values

Handle missing values in predictors

Encode categorical variables

Normalize / scale numerical features where required

Split dataset into *training* set and *validation* (unseen) set

Single-Model Development

Train *XGBoost* regression model on training data

Train Random Forest Regression (*RFR*) model on training data

Train Artificial Neural Network (*ANN*) model **using** *early stopping*

Tune hyperparameters **using** *cross-validation*

Evaluate each model using R^2 , *MAE*, *MSE*, and *RMSE*

Hybrid Model Development

Apply *K-Prototypes* clustering on *mixed-type* features

Assign cluster *labels* to training data

Train separate *RFR* models within each cluster

Aggregate cluster-based predictions into a *hybrid model* output

Evaluate *hybrid models* **using** the same performance metrics

Model Validation

Apply trained models to *validation* (unseen) dataset

Compare training and validation performance to assess *stability*

Identify best-performing model based on *accuracy* and *robustness*

Deployment and Visualization

Integrate final model outputs into a financial dashboard

Visualize revenue trends, predictions, and key financial indicators

Enable user-friendly interaction for business decision-making

End

Author Contribution – Conceptualization & Methodology: Um Hye Rin, Fathey Mohammed, and Narishah Mohamed Salleh – Data Curation & Analysis: Um Hye Rin, Mikkay Wong Ei Leen – Model Development & Implementation: Um Hye Rin, Fathey Mohammed, and Ibrahim T Nather Khasro – Dashboard Development & Integration: Um Hye Rin, Mikkay Wong Ei Leen, Ibrahim T Nather Khasro – Writing – Original Draft: Um Hye Rin, Fathey Mohammed – Writing – Review & Editing: Fathey Mohammed, Narishah Mohamed Salleh – Supervision & Project Administration: Fathey Mohammed, Narishah Mohamed Salleh.

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Data Availability Data is provided upon request.

Declarations

Competing Interests The authors declare no competing interests.

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